Run report: EXP-ENDO-7494e7 / RUN-ENDO-795989

Combined route: 2-GLV on a cheaper-model isogenous neighbour. This report records observations only. It makes no verdict; support, weakening, significance and priority belong to the evidence review.

Identity

- **Records:** experiment `EXP-ENDO-7494e7`, hypothesis `H-ENDO-f58749`, approval `DEC-20261016-00e9cd`, goal `GOAL-ENDO-001`. Source rows come from `RUN-ENDO-d7cb69` (sha256 `0576225d…`, checked by `run.py`).
- **Tasks:** executor `TASK-20261016-c886b3` (claimed, released `completed`); snapshot `TASK-20261016-d4363d`.
- **Run directory:** `experiments/EXP-ENDO-7494e7/runs/RUN-ENDO-795989/`.
- **Code commit** `c45d9a5969a19e7646511f6806f7ac0a02d972e4`. The manifest records `repository_dirty: true`: at launch, `harness/endosweep/isobench/` existed untracked. It is not imported by this run; independent validation `TASK-20261016-35d324` checks this.
- **Timing:** started 2026-10-16T19:42:21Z; wall time 74.7 s (measured); single process.

What ran

- **Command:** `tools/experiment_execution.py run` on the frozen plan, which runs `python -m harness.endosweep.isoglv` with these parameters:
  - `--ells 2,3,5,7,11,13 --depth 2 --max-nodes 64 --disc-bound 2000000`
  - `--build-limit 3 --build-seconds 900 --max-candidates 12 --verify-scalars 8 --seed 20261016`
  - two controls via `--include`.
- **Outcome:** trials 1 requested, 1 completed. Coverage `output_validated`. `check.py`: "ok: 9 rows, 0 with recorded errors".
- **Arms** (all modelled M_eq = M + S in `curvesweep`'s model):
  - **GLV on E:** `curvesweep.sweep_curve` on E.
  - **Model change:** copied from RUN-ENDO-d7cb69.
  - **Combined:** `sweep_curve` on the cheapest-conversion neighbour with a cheaper Weierstrass class on which the cheapest chain element builds and passes curvesweep's GLV-2 point check. Conversion (48 M_eq for one degree-5 step) is added for the one-shot column, and the route is verified with `isomodel.verify_route` on 8 scalars.
- **Derived gains:** each is 1 − arm / reference, computed from the recorded values.

Graph

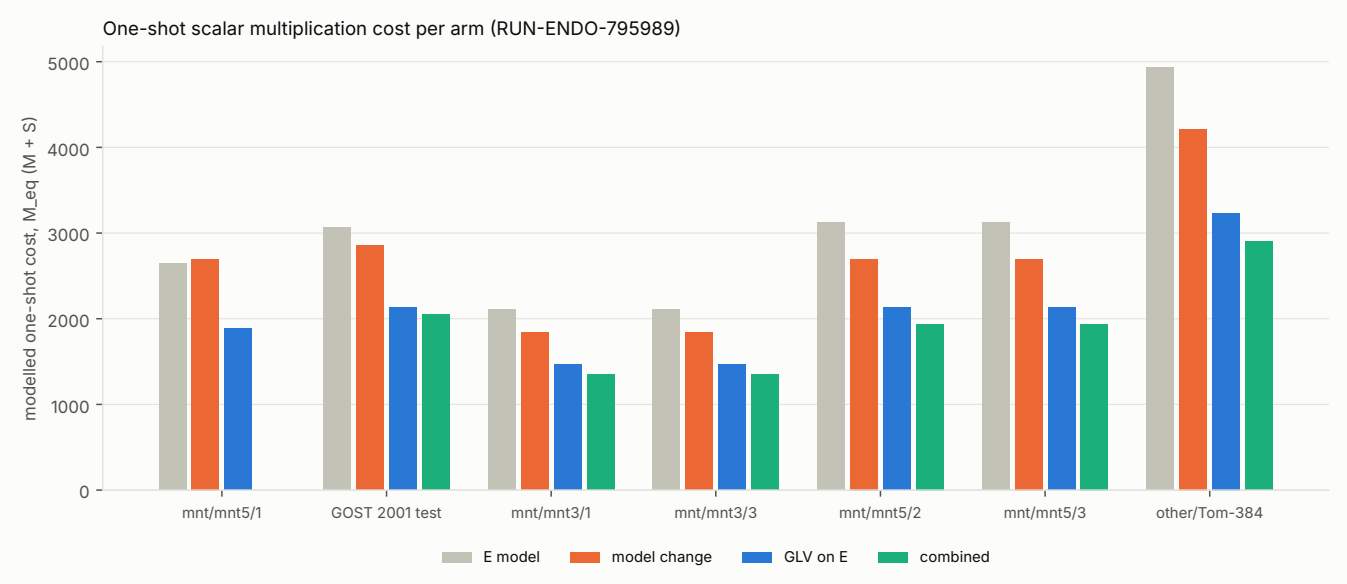


Figure 1: RUN-ENDO-795989, recorded rows only. One-shot modelled cost per arm on the six population curves with a small D_K, plus the negative control mnt/mnt5/1 (no cheaper neighbour, so no combined bar). Single deterministic run; no spread recorded.

Recorded values (one-shot unless stated)

curve	D_K	E model	model change	GLV on E	combined	attempts (path: built)	combined vs GLV on E, persistent	one-shot
GOST 2001 test	−915	3063.9	2858.7	2132.5	2053.5	5: yes	+5.96 %	+3.71 %
mnt/mnt3/1	−139	2103.7	1836.6	1466.5	1352.7	5: no, 5: no, 5: yes	+11.03 %	+7.76 %
mnt/mnt3/3	−139	2103.7	1836.6	1466.5	1352.7	5: no, 5: no, 5: yes	+11.03 %	+7.76 %

mnt/mnt5/2	−211	3124.4	2696.2	2131.4	1938.5	5: no, 5: no, 5: yes	+11.30 %	+9.05 %
mnt/mnt5/3	−211	3124.4	2696.2	2131.4	1938.5	5: no, 5: yes	+11.30 %	+9.05 %
other/Tom-384	−619	4929.5	4213.8	3233.8	2898.8	5: yes	+11.84 %	+10.36 %
mnt/mnt5/1 (control−)	−211	2648.2	2696.2	1890.5	–	none (0 candidates)	–	–
control+ ascent	−3	2796.5	2652.5	–	1885.8	11: yes	vs E model +36.43 %	+32.57 %
other/BADA55-VPR-224	none ≤ $2 \cdot 10^6$	2922.8	2588.4	–	–	not run (no GLV arm)	–	–

- The model-change column is RUN-ENDO-d7cb69′s `oneshot_M_eq`, so it includes its own conversion.
- The combined column is the neighbour's 2-GLV M_eq plus 48.0 M_eq of conversion.
- Recorded `model_change_vs_glv_on_E` (one-shot) on the six curves ranges from −25.2 % to −34.1 %.

Controls (recorded outcomes)

- **control+ascent:** the walk reached the j = 0 crater over one degree-11 step. Unit 2-GLV was priced there, and the route was verified on 8 scalars. Against E's own model: persistent +36.43 %, one-shot +32.57 %.
- **mnt/mnt5/1:** already isomorphic to an a = −3 model. It recorded 0 cheaper-class candidates and no combined route. Its own GLV on E was 1890.5.

Other observations

- On every small-D_K population curve, the combined route was found on a degree-5 neighbour and verified on 8 scalars.
- On four curves (mnt3/1, mnt3/3, mnt5/2, mnt5/3), one or two earlier degree-5 neighbours did not build the chain. In those four, 5 divides f_π.
- The neighbour's GLV cost equals the recorded GLV on E of the a = −3-isomorphic sibling in the same family: mnt3 1304.7 = mnt3/2′s figure in `research/endosweep_curves_20261006`, and mnt5 1890.5 = mnt5/1. This is an observation, not an identity check.

Missing or out of scope

- No timing in this run; the timed comparison is EXP-ENDO-33a1ee and covers BADA55-VPR-224 only.
- The conductor f_E of each neighbour is not certified.
- A failed chain build is recorded as an edge observation, not as evidence about End.
- Peak memory is not recorded.

Files

`plot.py` draws `arms.svg` from `runs/RUN-ENDO-795989/raw-result.json` only.